Continual Learning

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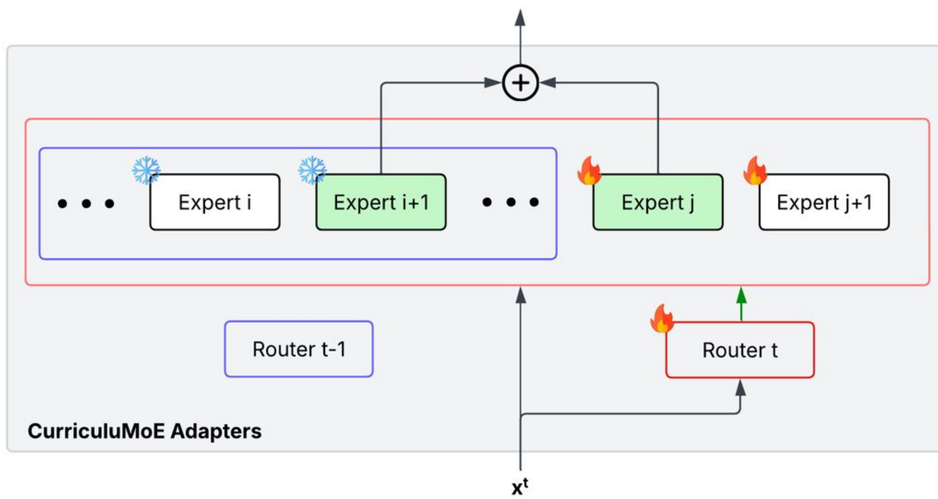
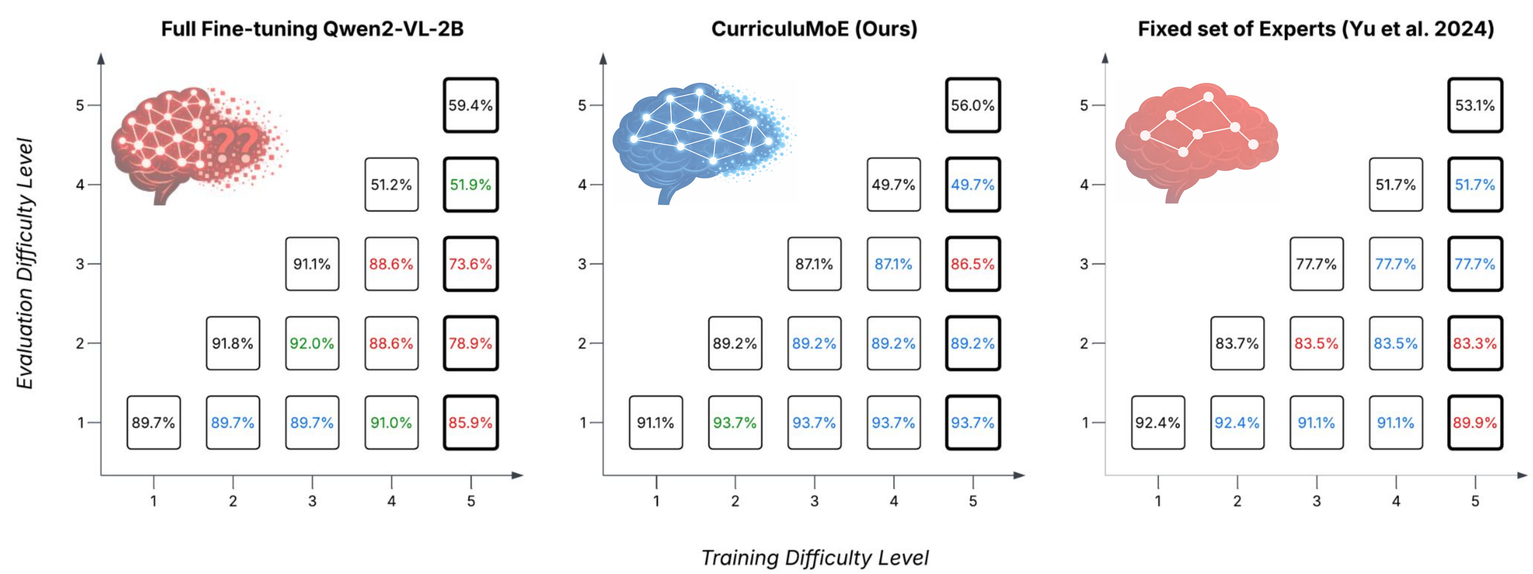
CurriculuMoE: Leveraging Task Compositionality to **Effectively Mitigate Catastrophic Forgetting in VLMs**

Summary

- VLMs suffer from **catastrophic forgetting** when continually learning new tasks
- We study how to mitigate this phenomenon in curriculum learning settings for VQA
- We propose **CurriculuMoE**, a Parameter-Efficient Fine-Tuning approach that **incrementally adapts new Experts from existing ones**
- We evaluate our approach on Spatial457, a synthetic benchmark for spatial reasoning in VQA
- CurriculuMoE strikes a **better balance between stability and plasticity** in curriculum learning settings than prior approaches, **mitigating catastrophic forgetting while efficiently learning new tasks**





Additional Results

- CurriculuMoE solves new tasks by effectively **balancing prior knowledge with task-specific expert updates**

Expert	L2	L3	L4	L5
Old Experts	65.0%	65.7%	70.5%	67.3%
New Experts	35.0%	34.3%	29.5%	32.7%

- CurriculuMoE achieves **near-oracle performance without ground-truth task labels** during inference

Task Label	L1	L2	L3	L4	L5
Oracle	91.1%	88.9%	87.9%	50.3%	54.9%
Classifier (Ours)	93.7%	89.2%	86.5%	49.7%	55.6%

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Examples and References available online!